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DNAm Deconvolution

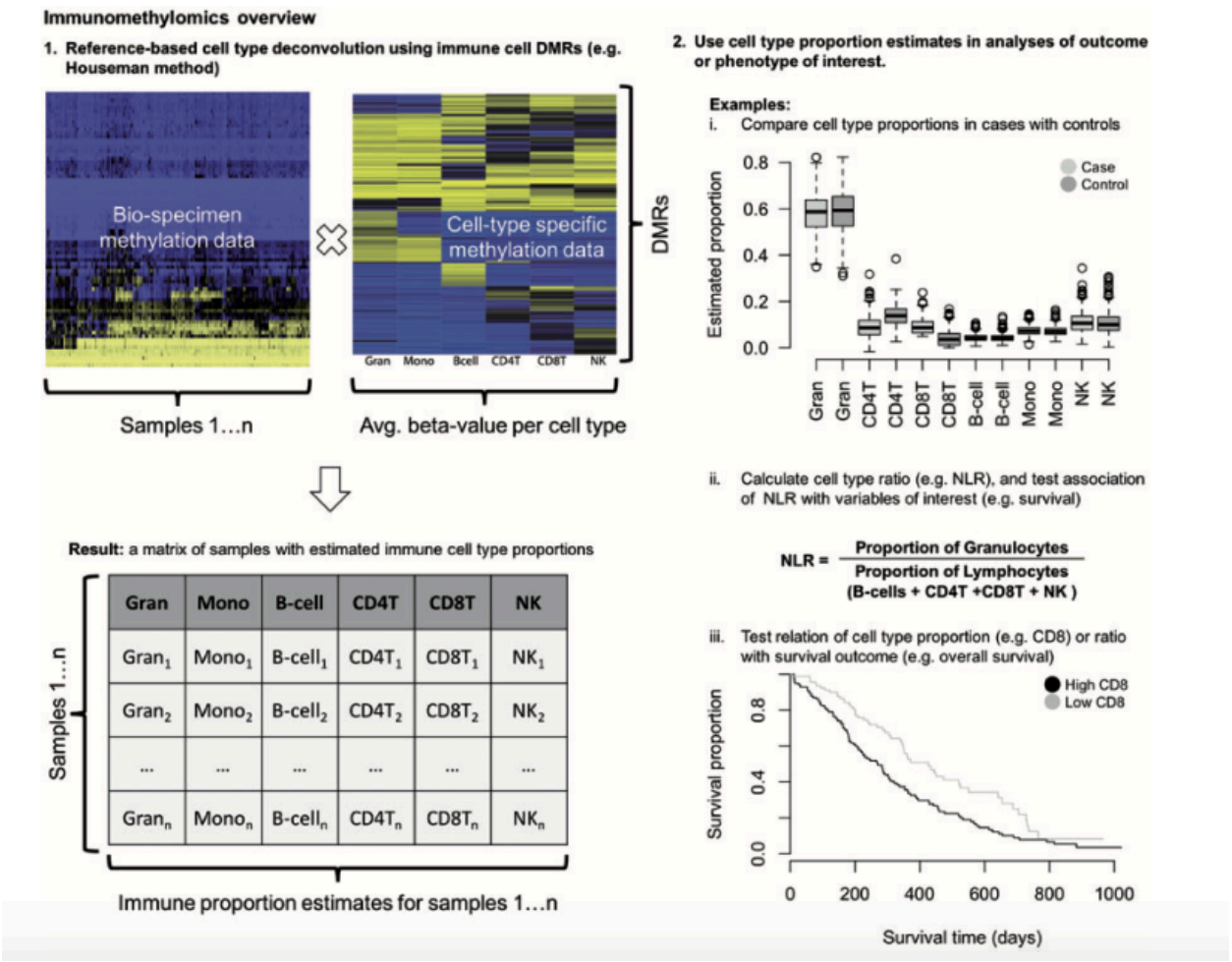
Motivation

infer immune cell proportions

- Inference of cell type proportions from DNAm also allows for tests of association between cell type proportions with various outcomes, phenotypes or exposures of interest across the spectrum of health and disease.
- EWAS that include adjustment for variation in cell type proportions. (cell type proportions can be included as covariates in regression models to adjust for potential confounding)
- The impetus behind developing and refining deconvolution algorithms comes from the confounding due to cell-type specific DNAm profiles. These signals may impede the discovery of DNAm changes associated with outcomes of interest.
- ***Comparison with gene expression-based deconvolutions***
  CellMix, CIBERSORT; VoCAL, ImmQuant – inferring immune cell-type composition from expression data.
  However, the number of transcripts may not be proportional to the number of cells; gene expression is more likely to be influenced by copy number variation than DNAm; RNA is less stable than DNA. → DNAm more accurate in cell proportion estimation!

- Cell proportions: previously, simple histological or flow cytometric assessments.
- Future directions: Improve reference-based, improve DMR selection / improve the subset of data used from the reference libraries.

- Test the relation of inferred proportions of immune cell types with the outcome of interest. Inference of CD4+ T-cell, CD8+ T-cell, B-cell, NK, Granulocyte and Monocyte proportions from DNAm using deconvolution has allowed us to directly use immune cell proportions in analyses.
  e.g. the neutrophil-to-lymphocyte ratio (NLR) is a consistent prognostic factor for survival in cancer, and can be derived using cell type proportion estimates from DNAm-based deconvolution of peripheral blood.



- (The most developed understanding of epigenetic markers of lineage commitment – immune cell subclasses defined by populations of distinct circulating blood cells).
  The composition of leukocyte populations – cancer states and toxicant exposures.
  (inform the underlying immune-biology of disease states and the immune response to almost all chronic medical conditions)

Methods

1. **Reference-based**

- In referenced-based deconvolution methods, the distribution of cell types is obtained by projecting whole-tissue DNA methylation data onto linear spaces spanned by cell-type-specific methylation profiles for a specific set of CpGs.
  (the CpGs that distinguish the cell types: differentially methylated positions (DMPs).)

--these methods require the existence of a **reference set** consisting of cell-type specific methylation profiles, such as those exist for blood. However, no such reference sets exist for solid tissues of interest, such as adipose and placenta, or even tumors. (motivates reference-free in next section)

$$Y = M\Omega^T$$

Y is m*n matrix representing DNA methylation data, n samples, m CpGs
M is a known m*K matrix, K cell types
Ω is n*K matrix representing subject-specific cell-type distributions

****
(reference-free methods attempt to circumvent lack of knowledge about M
Two-stage regression analysis or Zou approach: fitting a high-dimensional mixed effects model and equating the resulting random coefficients with cell-mixture effects.
Both rely on a predetermined model positing associations between DNA methylation Y and phenotypes X.

$$Y = AX^T + R$$

A and R is assumed to have further linear structure involving M and the common variation between A and R is assumed to represent systematic association with cell type distribution.

e.g.

$$Y = M\Omega^T$$

non-negative matrix factorization, the entries of the unknown matrices M and Ω are constrained to lie in the unit interval and the rows of Ω are constrained to sum to a value less than or equal to 1. However constrained factorizations can be computationally intensive and it's necessary to specify the number K cell types.
)

- (supervised methods that determine the underlying composition of cell types within a sample by leveraging <u>*differentially methylated regions (DMRs) specific to cell type*</u>, identified from DNAm measures of purified cell populations.)
- (constrained projection/ quadratic programming)
- (until recently, Houseman algorithm is the only algorithm!)
- Regression calibration: consider a methylation signature to be a high-dimensional multivariate surrogate for the distribution of white cells. In turn, this distribution is of interest for predicting/modeling disease states. Methylation signature – surrogate. (assumed to be highly correlated, yet imperfect, measure of leukocyte distribution. (measurement error models: the use of a noisy surrogate marker to investigate an association with a disease outcome of interest → biased estimates (internal or external validation data to "calibrate" the model and correct the bias)
- **Houseman** developed a method for cell type correction that capitalized on the idea that differentially methylated regions (DMRs) can serve as a signature for the distribution of different types of white blood cells. It uses these DMRs as a surrogate in a regression calibration based technique. – regression calibrations can lead to bias estimates and an

external validation data is used to calibrate the model and to correct for the bias. (Houseman: specifically for Illumina 27k beadchip array)

- **Jaffe and Irizarry**: adapt the Houseman method and tailored for Illumina 450k with 27k array.

Under most circumstances, the Houseman reference-based method implemented in the R package minfi is the optimal method for reference-based deconvolution.

2. **Reference-free**

- PCA forms the basis of many reference-free methods (SVA, ISVA, RUV, RefFreeEWAS, ReFACTor).
- Reference-free better than reference-based: can be applied to other tissues besides blood.
- (unsupervised methods for use when cell-type specific DMRs are not available.
- For reference-free deconvolution, SVA (Surrogate Variable Analysis) has the most robust sensitivity and specificity in the presence of high confounding.
- **FaST-LMM-EWASher**: applies the maximum likelihood in linear mixed models and optimizes spectral decomposition to estimate cell types.
- **RefFreeEWAS**: utilize SVD to decompose the residuals of unadjusted linear models along with unadjusted linear coefficient estimates, and estimates latent subject and cell-specific effects.
- **SVA**: Surrogate variable analysis estimates potential confounding factors from a SVD of residuals. (initially gene expression data); SVA decomposes the residual matrix and utilizes the permutations to identify statistically significant eigenvectors which serve as a representative of EH (eigengenes), and then infers surrogate variables based on these "eigengenes". – surrogate variables from SVA have the potential to cover information on cell types in DNA methylation from blood cells.
- **RefFreeCellMix**: improved over RefFreeEWAS. It uses a variant of non-negative matrix factorization to decompose the total methylation sites into CpG-specific methylation states for a pre-specified number of cell types and subject-specific cell-type distributions.
- **ReFACTor**: a variant form of PCA to adjust for the cell type effects. This method assumes that a small number of methylation sites are affected by underlying cell mixtures. It filters out CpGs if the variation is not large enough. It searches for top 500 most informative methylation sites and performs PCA with a fixed number of components on these CpG sites to obtain the components. These components can be used as a covariate in epigenome-wide association study or can be added one at time to remove the inflation due to cell type decomposition.
- Reference-free:
    1. Built upon linear mixed models with top principal components as covariates.
    2. infer latent variables for cell type compositions
    3. Surrogate variable analysis, utilize singular value decompositions and extract latent subject and cell-specific effects (surrogate variable analysis, based on SVDs of residuals in linear regressions, uses permutations to identify statistically significant eigen-vectors and consequently infer potential confounding factors (surrogate variables).

4. Remove unwanted variation, built upon factor analysis. Utilize reference CpGs inferred from a reference database, based on which factor analyses are conducted. The factors are then included in subsequent analyses for the purpose of adjusting for cell type effects.
5. Based on existing knowledge about the sparsity of informative genes, pathways and genetic functions.

Adjust for cell proportions

Several algorithms have been developed to measure and adjust for cellular heterogeneity in whole blood.
Whole blood is utilized in genome-wide association studies of DNA methylation patterns in relation to environmental exposures or clinical outcomes. These associations can be confounded by cellular heterogeneity. → cell type correction methods

We first evaluated the association of DNA methylation at each CpG across the whole genome with prenatal arsenic exposure levels and with cancer status, adjusted for estimated cell-type information obtained from different methods.

Via linear regression, we assessed the association of DNA methylation at each CpG site across the whole genome with prenatal urinary arsenic exposure levels, adjusting for cell-type effects with cell type inferred from one of the eight methods.
(The motivation of adjusting for cell types was due to the potential confounding effects of cell type compositions with respect to the association of arsenic exposure with methylation, caused by association of arsenic exposure with cell type compositions.

Real Data

Utilize a homogeneous real data set with some blood cells composition available and compared the true cell counts with the estimated cell compositions.
Three real data sets to compare the approaches

1. data on prenatal arsenic exposure and DNA methylation
2. an example data from FaST-LMM-EWASher
3. data on breast cancer status and DNA methylation

For each method, we recorded the number of CpGs showing statistically significant associations with prenatal urinary arsenic exposure after adjusting for multiple testing by controlling FDR at 0.05. Check the overlapping of identified CpGs and biological validity of identified CpGs. (e.g. one identified gene was not directly associated with arsenic exposure, but it has been known to regulate pacemaker activity in the heart and brain of mice and humans.)

check

- agreement in terms of identified CpGs potentially associated with a variable of interest.
- Agreement between the estimated cell type proportions and the physical counts of the cells.

Simulation

Simulated DNA methylation data under different settings with latent variables representing "cell types".

Scenario 1: we simulated DNA methylation data at 2000 CpG sites across 600 samples

- the first n CpG sites were associated with covariates of interest (e.g., level of arsenic exposure) and a set of latent variables
- the remaining CpG sites were only associated with the latent variables
- latent variables: cell types

1. One covariate ~ N(0,1), the coefficient of the covariate: 0.3, the intercept: 0.5
2. Five "latent" variables were used and generated from five different Normal distributions N(0,5), N(3,1), N(0,1), N(2,4), N(0,3). The association of DNA methylation and the latent variables was assumed linear and the coefficients were generated from N(0.5, 0.01).
3. The distribution of random errors in the linear regressions was N(0,1.2) for the n CpGs, N(0, 1.2) for the next 100 CpGs. N(0,2) for the remaining CpGs.
4. n=50, 100, 150 representing different sparsity levels (from high to low) of informative CpGs. 100 data sets for each n were simulated/
5. note that the covariates and latent variables have no correlation and are generated independently.

Scenario 2: latent variables have confounding effects. Overall setting same with Scenario 1. In addition, the covariate of interest and the five latent variables (6 variables in total) were correlated with correlations values.

Based on Copy number

(DNA copy number based)

ABSOLUTE: Absolute copy number inference

Inferring absolute copy number is more difficult for 3 reasons:

1. Cancer cells are nearly intermixed with an unknown fraction of normal cells (tumor purity)
2. The actual DNA content of cancer cells (ploidy), resulting from gross numerical and structural chromosomal abnormalities, is unknown.
3. The cancer cell population may be heterogenous, perhaps owing to ongoing subclonal evolution

In principle, one could infer absolute copy numbers by rescaling relative data on the basis of cytological measurements of DNA mass per cancer cell or single cell sequencing.

Purpose: infer absolute copy number per cancer cell from the population of mixed DNA

- Input: segmented copy-number data, together with precomputed models of recurrent cancer karyotypes and allelic fractions values for somatic point mutations

- Output: absolute copy number of local DNA segments and for point mutations, the number of mutated alleles.

α: cancer cells
$1-\alpha$: normal cells
$q(x)$: the integer copy number of locus x in cancer cells.
τ: the mean ploidy of cancer cell fraction (the average $q(x)$ across the genome)
Therefore, in the mixed sample,
the average absolute CN of locus x: $\alpha q(x) + 2(1-\alpha)$, the average ploidy of locus x: $D = \alpha\tau + 2(1-\alpha)$.
the average relative CN of locus x: $R(x) = \frac{\alpha q(x)+2(1-\alpha)}{D} = \frac{\alpha}{D}q(x) + \frac{2(1-\alpha)}{D}$

Based on Gene expression

1. Deconvolving the complete cellular composition
2. Assessing enrichments of individual cell types

Deconvolution (CIBERSORT)

Input: input matrix of reference gene expression signatures
(SVR: feature selection (adaptively selected to deconvolve a given mixture), robust to noise)

1. Design and validation of a leukocyte signature matrix
    - To assess the feasibility of leukocyte deconvolution from bulk tumors, we designed and validated a leukocyte gene signature matrix: LM22. (547 genes that distinguish 22 human hematopoietc cell phenotypes)
    - To validate the gene signature in LM22, we applied it to deconvolve external data sets of variably purified leukocyte subsets. CIBERSORT results matched ground-truth phenotypes in 93% of these data sets.
    - CIBERSORT – p value – sensitivity and specificity
      At p-value threshold of ~0.01, CIBERSORT achieved 94% sensitivity, 95% specificity (AUC = 0.98) for distinguishing positive from negative samples (scored the expression profiles as positive or negative depending on the presence/absence of at least one cell type in LM22.)
2. Performance on well-defined mixtures
3. Evaluation by simulation of bulk tissues
4. Deconvolution of closely related cell types
   Investigate CIBERSORT's discriminatory ability on cell types with correlated gene expression profiles. (Previous approaches avoid this issue by using cell type-specific marker gene or highly distinct GEPs.) In contrast, CIBERSORT doesn't require cell type-specific expression for every gene, suggesting applicability to diverse cell phenotypes
5. Summary: CIBERSORT exhibits substantially improved accuracy for the analysis of mixtures with (i) noise or unknown content (ii) closely related cell types. Drawbacks: Signature based methods: fidelity of reference profiles

Enrichment (ESTIMATE, xCell)

1. ESTIMATE

Filter stromal, immune signatures: ssGSEA – stromal, immune scores (to predict the level of infiltrating stromal and immune cells – tumor purity)
Estimate the association of stromal/immune scores with tumor purity.
Compare ESTIMATE scores with predictions based on ABSOLUTE (establishes the fraction of tumor cells based on somatic DNA copy number alterations.
From data, significant correlation between stromal, immune scores (ESTIMATE) and ABSOLUTE purity prediction.

2. xCell

Using the data sources and based on different thresholds, we derive gene signatures for 64 cell types.
Raw enrichment score per cell type: the average ssGSEA score of all signatures corresponding to the cell type.

Primary goal: accurately identify enrichment of cell types in mixtures.
Simulations: performed an array of simulations of gene expression combinations for different cell types to assess the accuracy and sensitivity of gene signatures.

- Our raw scores reliably predict even small changes in proportion of cell types, distinguish between most cell types and are reliable in different transcriptomic analysis platforms.
- However, the raw score of RNA-seq samples are not linearly associated with cell type abundances: they do not allow comparisons across cell types.
- By using the training samples we generated using synthetic expression profiles by mixing the cell type of interest with other non-related cell types – fit a transformed score – comparison across cell types
- Adjusted score: using the reference training data sets, for each cell type we generated random mixtures of all cell types except the corresponding cell type and calculate the cell-type adjusted scores. We then fit a beta distribution for each of cell types and used these distributions to calculate the probability that the score of corresponding cell type is present in the mixture by random.

Pan-cancer methyl-based deconvolution

The nature and extent of immune cell infiltration into solid tumors are key determinants of therapeutic response.

Emerging evidence from clinical trials indicates higher response rates in those cancer types that display greater lymphocyte infiltration.

Although flow cytometry of disaggregated tumor biopsies is commonly used for investigating cellular composition, this is often unfeasible for several reasons: difficulty in obtaining fresh

tumor tissues; lack of defined markers for poorly characterized cell types; high cost of labour, reagents and equipment

CIBERSORT and methylation data

CIBERSORT: support vector regression modeling to gene expression microarray data

DNA methylation data are also suitable for deconvolution of tissue mixtures
(Studies so far have focused primarily on simple tissues such as blood, where cell type differences are a major confounder in EWAS)

MethylCIBERSORT
To develop a methylation-based deconvolution pipeline for application in tumors, we created a custom R interface to produce basis matrices for use with CIBERSORT and generated a reference using fibroblasts and seven different immune cell types.
Combining methylation-based feature selection from both stromal and cancer cells with the robust performance of CIBERSORT previously displayed on gene expression microarray data allowed us to derive estimates for different infiltrating cell populations as a fraction of the overall sample.

Validation of DNA methylation-based deconvolution for the analysis of tumor compositions

1. Correlation between MethylCIBERSORT fractions and flow cytometry for PBMC mixtures
2. Spearman's correlation between ABSOLUTE and MethylCIBERSORT versus other previously published purity estimation methods.

An important potential advantage of DNA-methylation over gene expression-based deconvolution methods is the ease with which accurate DNA methylation profiles can be obtained from formalin-fixed, paraffin-embedded (FFPE) samples.

SVA (reference-free, gene expression)

Sources of signal due to factors that are unknown, unmeasured or too complicated to capture through simple models.
Failing to incorporate these sources of heterogeneity into analysis → problems
Solution: introduce SVA to overcome the problems caused by heterogeneity in expression studies. (recover the effects of important missed variables);
SVA pros: adjusting for surrogate variables reduces differential expression due to sources other than primary variables; SVA orders the significant gene lists to more accurately and reproducibly reflect the orderings of the genes with respect to their true differential expression signal; improves accuracy and consistency in detecting differential expression; SVA capatures the expression heterogeneity patterns to improve the characterization of biological signal in expression analyses.

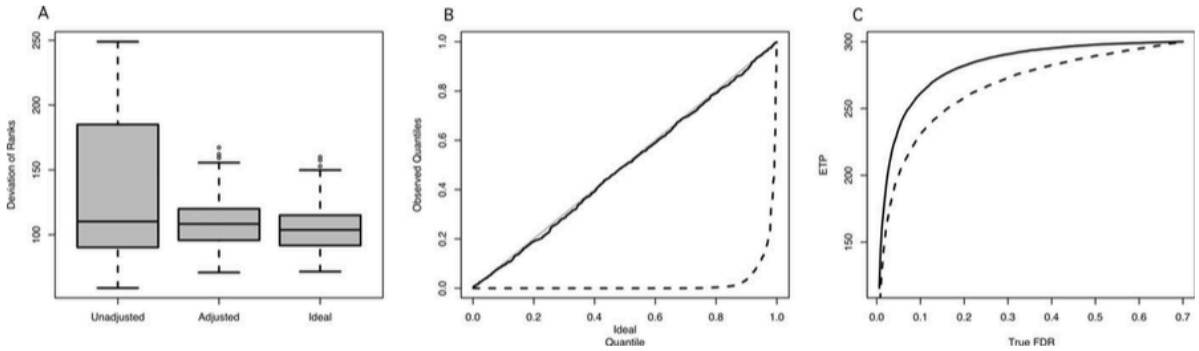


Figure 1. Impact of Expression Heterogeneity

One thousand gene expression datasets containing EH were simulated, tested, and ranked for differential expression as detailed in Simulated Examples.
(A) A boxplot of the standard deviation of the ranks of each gene for differential expression over repeated simulated studies. Results are shown for analyses that ignore expression heterogeneity (Unadjusted), take expression heterogeneity into account by SVA (Adjusted), and for simulated data unaffected by expression heterogeneity (Ideal).
(B) For each simulated dataset, a Kolmogorov-Smirnov test was employed to assess whether the *p*-values of null genes followed the correct null Uniform distribution (Text S1). A quantile–quantile plot of the 1,000 Kolmogorov-Smirnov *p*-values are shown for the SVA-adjusted analysis (solid line) and the unadjusted analysis (dashed line). It can be seen that the SVA-adjusted analysis provides correctly distributed null *p*-values, whereas the unadjusted analysis does not due to EH.
(C) A plot of expected true positives versus FDR for the SVA-adjusted (solid) and -unadjusted (dashed) analyses. The SVA-adjusted analysis shows increased power to detect true differential expression.
doi:10.1371/journal.pgen.0030161.g001

Background: large-scale gene expression studies allow one to characterize transcriptional variation with respect to measured variables of interest, such as differing environments, treatments, time points, phenotypes or clinical outcomes.

Sources of heterogeneity:

1. Unmodeled factors: sources of variation explained by measured variables, but are not explicitly included in the statistical model (e.g. because their relationship to expression is intractable or the relevant measured variables were excluded because of sample size restrictions)
2. Unmeasured factors: sources of expression variation that are not measured in the course of the study
3. Gene-specific noise: random fluctuations in gene expression independently realized from gene to gene.

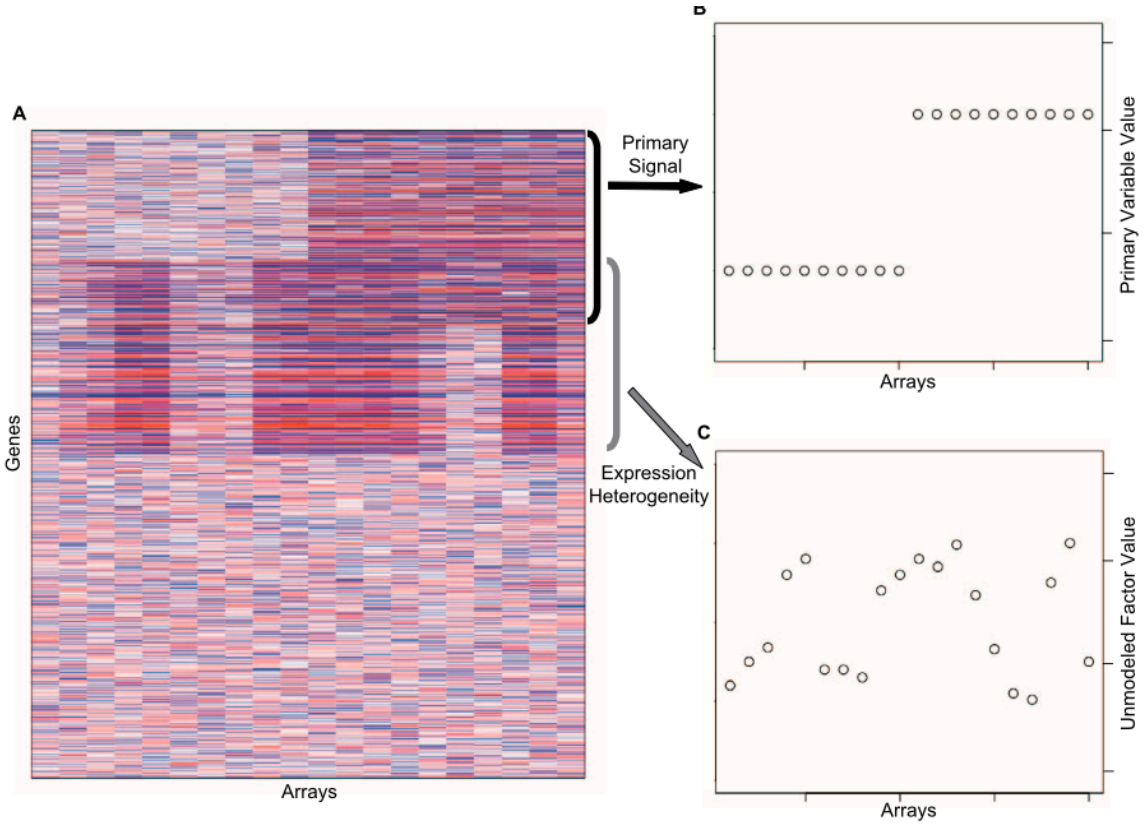


Figure 2. Example of Expression Heterogeneity

(A) A heatmap of a simulated microarray study consisting of 1,000 genes measured on 20 arrays.
(B) Genes 1–300 in this simulated study are differentially expressed between two hypothetical treatment groups; here the two groups are shown as an indicator variable for each array.
(C) Genes 201–500 in each simulated study are affected by an independent factor that causes EH. This factor is distinct from, but possibly correlated with, the group variable. Here, the factor is shown as a quantitative variable, but it could also be an indicator variable or some linear or nonlinear function of the covariates.

SVA: borrows information across genes to estimate the large-scale effects of all unmodeled factors directly from the expression data. After the surrogate variables are constructed, they are then incorporated into any subsequent analysis as covariates in the usual way.
Step 1: remove the signal due to the primary variables of interest to obtain a residual expression matrix. Apply SVD to the residual expression matrix to identify signatures of expression heterogeneity in terms of an orthogonal basis of singular vectors that completely reproduces these signatures. Use a statistical test to determine the singular vectors that represent significantly more variation than would be expected by chance.
Step 2: identify the subset of genes driving each orthogonal signature of EH through a significant analysis of associations between the genes and the EH signatures on the residual expression matrix
Step 3: for each subset of genes, build a surrogate variable based on the full EH signature of that subset in the original expression data
Step 4: include all significant surrogate variable as covariates in subsequent regression analyses, allowing for gene-specific coefficients for each surrogate variable

Houseman (ref-based, DMRs, methyl)

DNA methylation arrays as surrogate measures of cell mixture distribution

Background: Leukocyte composition will critically inform the underlying immuno-biology to most chronic medical conditions; emerging research shows that DNA methylation is responsible for cellular differentiation. The composition of leukocyte populations is well known to reflect disease states and toxicant exposures and can be altered by signaling cascades that prompt migration of whole classes of cells into or out of tissues.

Here, we present a method (similar to regression calibration) for inferring changes in the distribution of white blood cells between different subpopulations using DNA methylation signatures, in combination with a previously obtained external validation set consisting of signatures from purified leukocyte samples. (inferring cell proportion changes)

DMRs: individual assays identifying cell-specific DMRs have proven useful for quantifying individual cell types in human tissues and peripheral blood. However, these assays are limited to detecting the relative proportion of one individual cell type compared with all others. Hence, an approach that allows for the simultaneous quantification of the entire distribution of cell types, using an array of biomarkers based on generally available technology, would be considerably more informative, especially in studies of human disease and exposures.

The backbone of the approach is the DNA methylation signature of each of the principal immune components of whole blood. We essentially seek a form of regression calibration, where we consider a methylation signature to be a high-dimensional multivariate surrogate for the distribution of white blood cells. As a surrogate, the DNA methylation signature is assumed to be a highly correlated, yet imperfect, measure of leukocyte distribution, and thus fits into the framework of measurement error models.

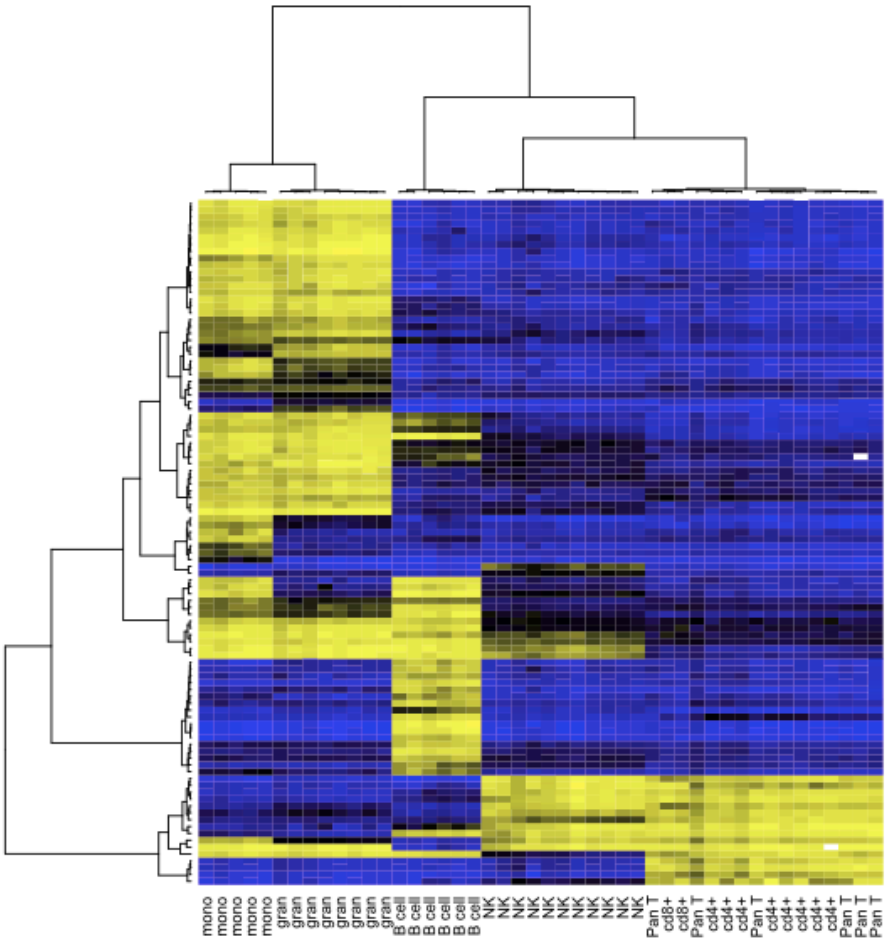


Figure 1 Clustering heatmap for external validation white blood cell data (S_0). Yellow = unmethylated ($Y_{hj} = 0$), black = partially methylated ($Y_{hj} = 0.5$), blue = methylated ($Y_{hj} = 1$).

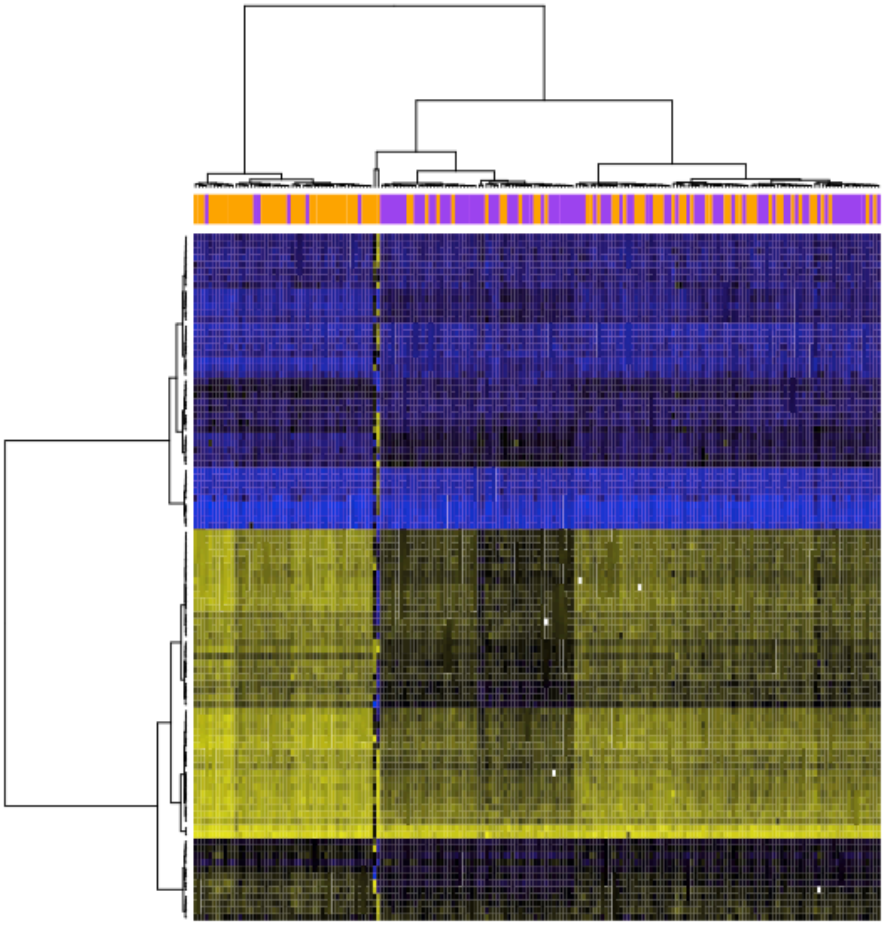


Figure 2 Clustering heatmap for target HNSCC data (S_1). Yellow = unmethylated $(Y_{ij} = 0)$, black = partially methylated $(Y_{ij} = 0.5)$, blue = methylated $(Y_{ij} = 1)$. The annotation track above the heatmap indicates case-control status (orange = case, purple = control).

types not profiled). In every simulation, we specified S_0 to consist of 5 B-cell samples, 10 granulocyte samples, 5 monocyte samples, 15 NK samples, 5 general "Pan-T" T-cell samples, 8 specific CD4+ T cell samples, and 2 specific CD8+ T cell samples. Estimates from the external validation set S_0, described above, were used for mean methylation profiles among WBC types, using the $m = 100$ most informative CpG sites.

We specified $n_1/2$ cases and $n_0/2$ controls, $n_0 \in \{100, 200, 500\}$. Among the controls, methylation profiles were generated by a white blood cell population of 7% B-cells, 62% granulocytes, 6% monocytes, 2% NK cells, and 13% were T-cells, of which 65% were CD4+ cells and 35% were CD8+ cells, and the remaining 5% were unspecified (and assumed to have mean methylation equal to that of the unsorted T-lymphocytes). Among cases, we specified one of the following scenarios: a 4% reduction in CD4+ cells, a 2% reduction in CD8+ cells, and an 8% increase in granulocytes (alternative with changes in both CD4+ and CD8+, "Strong Alternative I"); a 6% reduction in CD4+ cells, and an 8% increase in granulocytes (alternative with changes in CD4+ but not CD8+, "Strong Alternative II"); a weaker alternative with half the effects of Strong Alternative I ("Mixed Alternative" elaborated upon below); and

Conclusions

The method we present here has potentially far reaching implications for rapid, simple and complete assessment of the composition of human white blood cell populations, i.e. the immune profile. Currently, assessment of the cellular composition of peripheral blood cannot be accomplished without the use of freshly drawn venous blood that is immediately prepared in a specially equipped laboratory. A complete assessment of the entire immune profile requires extensive flow cytometric measurements based on protein epitopes on leukocyte membranes that distinguishes subtypes of immune cells that are either too rare or too similar in appearance to be distinguished using simple microscopic approaches. In particular, flow cytometry is limited by the following: (i) cells must be separated, requiring large volumes of fresh cells; (ii) detection can be accomplished only by the fluorescent antibody tags available, which require expensive technology to read; (iii) the outer cell membrane must be intact, mandating limited utility in many instances (particularly in research). In contrast, our method requires the application of these labor-intensive or expensive steps only in the construction of the validation set S_0, which need only be developed once. Once S_0 is available, subsequent interrogation is based on the chemically stable CpG methylation of DNA; thus our method obviates the need for fresh blood and the preservation of labile protein epitopes. It is also able to simultaneously assess all of the individual components of the peripheral blood using a highly multiplexed molecular platform and is thus very straightforward logistically. Furthermore, the statistical methodology presented here can be implemented easily with the instrumental output of the methylation arrays, which simplifies the interpretation of the immune profile data from the operators point of view. This method can be immediately deployed in a research framework to cost effectively assess human immune profiles (in fresh or archival samples), exploring their potential as biomarkers, and addressing key questions regarding disease pathogenesis. Furthermore, our approach is readily suited for rapid translation to a broad base of clinical applications such as disease monitoring, diagnosis, prognosis, and response to therapy.

Our approach makes research on biobanked specimens possible, now making a vast array of prospective studies that could not otherwise be done, possible. Software and sample data are provided in Additional file 2.

ISVA (methylation, mRNA expression)

Confounding problem:
A common problem in large-scale microarray profiling studies. For example, a study profiling hundreds of samples over a relatively large number of chips is vulnerable to confounding by chip effects. Even if the study is well-designed and balanced in relation to the phenotype of interest, logistical or sample quality issues may induce confounding. Moreover, inter-chip variation cannot always be corrected for, even with powerful inter-array normalization techniques such as quantile normalization. (Bolstad *et al.*, 2003; Teschendorff *et al.*, 2009). Global normalization methods such as quantile normalization may also not be desirable if they obscure biologically important global differences between samples, as it may happen for example in DNA methylation studies (Laird, 2010; Moore *et al.*, 2008). A further difficulty is that often the confounding factors (CFs) are unknown to the experimentalist (Leek and Storey, 2007) or, if known, may be subject to substantial uncertainty or measurement error. In these scenarios, techniques that have been used to explicitly adjust for known and error-free confounding factors (Johnson *et al.*, 2007) cannot be used.

In large-scale studies using Illumina Infinium Human Methylation 27k array, we generally observe two main cofounders: beadchip effects and bisulfite conversion efficiency. (variation due to these cannot removed by quantile normalization procedures)

Variation in dataset has 3 parts:

1. One part reflects the biologically interesting variation
2. Another reflects technical 'unwanted' variation caused by known or hidden confounders
3. The 3rd part is the inherent stochastic (random) variation.

A common difficulty in large-scale microarray studies is the presence of confounding factors (problems: skew estimates of statistical significance, cause unreliable feature selection and high false negative rates) → solution: SVA

(i.e. SVA – to identify features correlating with a phenotype of interest in the presence of potential confounding factors)

Using simulated data, we show that SVA:

1. Identify confounders
2. Adjust for confounding
3. A useful feature selection tool in studies that are subject to confounding.

FDR and correlations.

SVA: The main concept behind SVA is to model the potential confounding factors as singular vectors ('surrogate variables') derived from SVD.

ISVA: we propose that confounding factors would be better modeled as statistically independent surrogate variables.
E.g. two confounders such as beadchip and age are statistically independent variables. This requires the variables to be uncorrelated in a non-linear fashion, a stronger condition that the linear uncorrelatedness imposed by an SVD.
In order to model CFs as statistically independent variables, we propose to use ICA in a framework similar to SVA. – ISVA
(ICA vs PCA/SVD: better in identifying alterations in biological pathways and regulatory modules.)

ISVA vs SVA:

1. The use of ICA instead of PCA to identify variables that represent potential confounders
2. Inclusion of a surrogate variable selection step

Discussion and conclusions:
We have proposed an extension of the SVA framework which relies on ICA to model potential confounders and to subsequently use these as covariates for feature selection.
Results across methylation and mRNA expression show:

ISVA improves the identifiability of CFs over the SVD/PCA framework used in SVA (more robust algorithm for feature selection and downstream statistical inference)

SVA was never intended as an algorithm to identify confounders, but rather only as a tool for finding a basis for the relevant subspace. Thus, while ISVA may improve the identifiability of confounders, the subspace spanned by these might be the same as that spanned in SVA. In this scenario, SVA and ISVA would therefore lead to very similar feature selection.

ISVA is useful for deriving lists of differentially altered features in the presence of known or hidden confounding factors. (perform joint modeling of the phenotype and confounders)

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van 't Veer,L.J. *et al.* (2002) Gene expression profiling predicts clinical outcome of breast cancer. *Nature*, **415**, 530–536.
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As with SVA, there are two parts to the algorithm: (i) detection of confounding/unmodeled factors (steps 1–4) and (ii) construction of independent surrogate variables (ISVs) (steps 5–10):

(1) Construction of the residual variation matrix by removing the variation associated with the phenotype of interest: $R_{ij} \equiv X_{ij} - \hat{f}_i(y_j)$.

(2) We estimate the intrinsic dimensionality, K, of the residual variation matrix using Random Matrix Theory. This gives the number of components as input to the ICA algorithm.

(3) Perform ICA on R: $R = SA + \epsilon$, with S a $p \times K$ and A a $K \times n$ estimated data matrix.

(4) The significant independent components are given by the columns of S (S_k) and rows of A (A_k), respectively.

(5) We regress A_k to each X_i ($i = 1, \ldots, p$) and calculate P-values of association p_i.

(6) From this P-value distribution, we estimate the FDR using the q-value method (Storey and Tibshirani, 2003) and select the features with $q < 0.05$. If the number of selected features is less than 500, we select the top 500 features (based on P-values). Let r_k denote the number of selected features.

(7) We construct the reduced $r_k \times n$ data matrix X_r obtained by selecting the features in previous step.

(8) Perform ICA on X_r using K independent components: $X_r = S_r A_r + \epsilon_r$. Find the column k^* of A_r that best correlates (absolute correlation) with A_k.

(9) Set the ISV $v_k = (A_r)_{k^*}$.

(10) Repeat steps 5–9 for each significant independent component, A_k, obtained in step 4.

To finally identify features correlating with the phenotype of interest, we adjust for the ISVs using the general model:

$$X_{ij} = f_i(y_j) + \sum_{k=1}^{K} \lambda_{ki} v_{kj} + \epsilon_{ij} \qquad (6)$$

Methylation data:
the normalized data as described in (Teschendorff *et al.*, 2010)

methyl 450k ref-based

Accounting for cellular heterogeneity is critical in EWAS

The cell composition variability in epigenetic studies based on whole blood and other heterogeneous tissue sources.
Existing cohort studies that extensively characterize participants often store blood samples. The most widely available tissue for subsequent/retrospective EWAS is whole blood. Observed whole blood DNAm profiles are therefore mixtures of the cell type profiles.
Cellular composition explains much of the observed variability in DNAm.

Houseman, ref-based: describe a statistical method that can accurately estimate relative proportions of cell type components in whole blood.
(similar statistical approach)
Guintivano: a method for estimating neuron and non-neuron components in brain samples

Primary interest: age effects
Confounders: cell type
Tissue: whole blood DNA methylation

Confounding between cell composition and age → false positives

Adjust for:
Due to high levels of confounding, we currently do not recommend regression approached for adjustment. → RUV performed best for reducing the composition-based confounding

Data: There were five publicly available datasets on the Illumina 450k platform [5,8,10,11,26] performed on blood samples in the Gene Expression Omnibus (GEO) available through the National Center for Biotechnology In- formation (NCBI) as of February 2013

Ref-based blood

Background – blood:
Given the limitations to obtain large number of samples from affected tissues, blood is an attractive, easy and available source of DNA. Studies suggest that DNA methylation differences can be detected in the blood of patients with cancer, even for solid tumors.
** there is increase study comparing differences in DNA methylations in whole blood between cases and controls for complex disease:

25. Chowdhury S, Erickson SW, MacLeod SL, Cleves MA, Hu P, et al. (2011) Maternal genome-wide DNA methylation patterns and congenital heart defects. PLoS One 6: e16506.
26. Nadeau K, McDonald-Hyman C, Noth EM, Pratt B, Hammond SK, et al. (2010) Ambient air pollution impairs regulatory T-cell function in asthma. J Allergy Clin Immunol 126: 845–852 e810.
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28. Lin SY, Hsieh SC, Lin YC, Lee CN, Tsai MH, et al. (2012) A whole genome methylation analysis of systemic lupus erythematosus: hypomethylation of the IL10 and IL1R2 promoters is associated with disease activity. Genes Immun 13:214–20.

Methyl Brain mixture ref-based

Confounding: Brain cellular heterogeneity may bias DNA methylation patterns, influencing findings in psychiatric epigenetic studies; control of cellular heterogeneity will enable more robust hypothesis testing in the brain.

We performed fluorescence activated cell sorting (FACS) of neuronal nuclei and Illumina HM450 DNA methylation profiling in post mortem frontal cortex of 29 major depression subjects and 29 matched controls.

R package: CETS – quantify neuronal proportions; generate in silico neuronal profiles capable of removing cell type heterogeneity bias from DNA methylation data.

CETs derived neuronal proportions correlated significantly with age in the frontal cortex and cerebellum and accounted for epigenetic variation between brain regions. / can transform data to eliminate (or at least reduce) the confounding effect of cell type heterogeneity on methylation profiles.

RefFreeEWAS (methyl, SVA related)

Methods infer changes in cell mixture proportions; adjust for cell mixture effect in the context of EWAS.

However, these adjustments require existence of reference datasets, which may be expensive to collect. For some tissues such as placenta, saliva, adipose or tumor tissue, the relevant underlying cell types may not be known.

Recall – ref-based:

These methods may broadly be conceived as projection of DNA methylation data from a target dataset S1 onto a reference dataset S0 consisting of DNA methylation profiles for isolated cell types. For example, S1 may consist of DNA methylation measured in whole blood from a case-control study, while S0 may consist of corresponding DNA methylation profiles from isolated leukocyte cell types.

RefFreeEWAS vs SVA:
RefFreeEWAS (closely related with SVA) relies on a simple projection based on SVD, as does SVA.
In SVA, the residuals of a linear model are decomposed into a factor-analytic structure and the factors are used subsequently in a regression model, with iteration resulting in a final set of surrogate variables.
RefFreeEWAS , which doesn't require iteration to obtain estimates, includes unadjusted linear coefficient estimates as columns of the matrix to be decomposed. This construction associates the residuals of the unadjusted model with the unadjusted coefficient estimates in a manner consistent with a linear mixing assumption.

RefFreeEWAS:

- This method posits a statistical model that involves a latent variable representing mean methylation, together with a factor-analytic error model consistent with SVA and ISVA.
- Our approach now offers the possibility of conducting EWAS adjusting for mediation by cell type even without the existence of reference datasets.
- Unlike SVA and ISVA, we posit a specific data generation model that incorporates a factor-analytic structure that is implicit in the previously published methods. However, we incorporate all CpG features in the factor-analytic structure, rather than attempting to select a subset of features that are optimally informative.

ReFACTor (methyl, ref-free, PCA)

Background:
The standard statistical analysis applied in EWAS uses a univariate test for correlation between the phenotype and each of CpG sites.
In a standard EWAS of primary tissue, such as whole blood, methylation data represent the epigenetic states of a heterogeneous mixture of cell types.
Confounding: Correlations between the phenotype of interest and the cell type composition lead to large number of false discoveries.

The false discoveries due to cell type heterogeneity can be addressed by adding the cell proportions as covariates.
→ **reference-based** to estimate the cell proportions:
Using a reference data set that includes methylation measurements for sorted cells.
(problem: <u>only for blood cells</u>! Reference data for whole-genome methylation levels from sorted cells are available only for a small subset of different blood cells and not for any other tissues.

Furthermore, the existing data sets are small, and the individuals in the reference data are not matched for methylation-altering factors such as age and sex, which may lead to inaccurate cell type estimates.)

reference-free

ReFACTor can be applied to any tissue.

It provides improved estimates of cell type composition, resulting in improved power and control for false positives in EWAS.

Real data:

Unsupervised methods for cell type correction, using the first several PCs of a standard PCA, FAST-LMM-EWasher, RefFreeEWAS.

Evaluate the correlation between measured cell type proportions and inferred ReFACTor components.

Simulation

Compare 5 approaches: uncorrected linear regression, linear regression with PCA, linear regression with ReFACTor components, FAST-LMM-EWasher, RefFreeEWAS

1. Evaluate the control for false positives
   We found that none of these methods adequately controls for false positives, however, ReFACTor obtains a significantly reduced level of false discoveries.
2. Compare the fraction of simulated dataset in which the truly associated methylation site obtained the best P value (the detection power)

Algorithm

Fast-LMM-EWASher (methyl, ref-free, Zou, adjust but no proportion, TCGA)

Background:

EWAS

The goal of EWAS is to identify changes in the epigenome at particular loci that are correlated with a phenotype of interest.

Confounding

Facing many of the same challenges as GWAS, EWAS additionally face the major challenge that the epigenome can be highly variable across different cell types. This heterogeneity, in conjunction with the fact that cases and controls typically differ in their cell-type composition, results in spurious associations – those that only tag the cell type rather than reveal more fundamental biological signal of interest.

Bottleneck

Obtain accurate estimates of the cell-type composition for each sample in the study and to then use these estimates as covariates.

By pass the bottleneck:

Fast-LMM-EWASher

It automatically corrects for cell-type composition without the need for explicit knowledge of it.
Removes spurious associations arising from cell-type heterogeneity, while yielding true associations of interest, without any knowledge of cell-type composition.
Linear mixed model + PCA!

- we compute the methylome similarity between every pair of samples and then use these similarities as the covariance in the mixed model as an implicit proxy for cell-type composition
- **simulation**: using the empirical distribution of cell-type composition for the cases and controls in the RA data set, we simulated methylation profiles for a set of cases and controls that had no true associations with the case-controls status. On the null-only simulated data, FaST-LMM-EWASher was well calibrated (yielded noninflated P values) and didn't yield any false positive loci passing the Bonferroni threshold.
- **TCGA data:** solid tumor-based data for breast and colon cancer. (solid tumor-based data are likely to contain heterogeneous mixtures of many cell types – many of which may be unknown and most of which do not have reliable reference methylation profiles.)

RUV (ref-free, methyl)

SUMMARY: Deconvolution – EWAS (ref-based/ref-free, +hierarchical)

It's unclear how best to adjust for cell-type heterogeneity, as this may also largely depend on the type of tissue and phenotype being considered.

Once the proportions of the underlying cell types in the sample have been estimated, these can then be incorporated as covariates in supervised models. This then allows identification of DMPs, where the differential methylation is not driven by underlying changes in cell-type composition.

1. Ref-based:

Use a priori defined set of DNAm reference profiles for the cell types of interest. These reference profiles are defined over a selected set of CpGs, typically those which maximize the discriminative power of the different cell types. Given reference DNAm profiles, the method proceeds by postulating that a sample's DNAm profile is a weighted linear sum of these reference profiles. One then attempts to find the weights which best explain the observed sample profile, in other words, methods typically use a least-squares minimization procedure to accomplish this.
** So far, these reference DNAm profiles have been generated for tissues such as whole blood (peripheral blood mononuclear cells PBMC), cord blood, prefrontal cortex and breast.

Houseman – Constrained projection/Quadratic programming

Briefly, in CP/QP inference proceeds via least-squares minimization but subject to the constraint that weights cannot be negative and that the sum of weights must add to 1, or alternatively, that the sum adds to a number which is less than or equal to 1. (the latter constraint allowing for the possibility that the reference database doesn't contain all relevant cell subtypes.)

CP vs CIBERSORT

Houseman's CP algorithm has been extensively tested and demonstrated to work extremely well on whole blood. However, it is at present unclear whether CP/QP technique is the best possible algorithmic framework among reference-based methods.
*** for instance, in the context of gene expression, CIBERSORT (a nonconstrained reference-based) was found to be better than CP/QP. (reason: CP/QP may have the normalization issues).

CIBERSORT – uses an advanced form of Support Vector Regression to perform what is effectively a penalized multivariate regression, imposing non-negativity and normalization constraints only a posteriori, in other words, after estimating the regression weights. (this gives CIBERSORT more flexibility to search for a wider solution space, yet it may also be more susceptible to large sources of noise)

(CP algorithm is only optimal when noise levels are random and very large.
For realistic noise levels, robust partial correlations and CIBERSORT are better than Houseman's CP.
While these results indicate that CP/QP may not be the optimal framework for inference, reference-based algorithms remain underexplored in the context of EWAS, pointing toward the critical need to develop improved forms of existing algorithms.)

Constructing reference database (quality):

The quality of reference DNAm profiles impacts on the quality of subsequent inference

Two related aspects of reference profiles that need particular attention are the number of features (i.e. CpGs/probes) and the resulting collinearity of cell-type reference profiles.

Highly colinear reference profiles can cause well-known difficulties for statistical inference.

For this reason, optimization of reference database (i.e. selection and number of CpGs) through optimization of the conditioning number of the reference DNAm matrix, may result in a reference DNAm database that is more stable to random perturbation, for instance, to random removal of CpGs, as it may happen in practice when not all CpGs in the reference are available in the sample of interest.

Another potential problem with constructing a reference DNAm database is the use of reference samples that are not representative of the cell types encountered in real EWAS. For instance, genetics and, in the case of primary cells, the age of the donor tissue can impact on DNAm.

Another possibility to improve the quality of a reference DNAm database is to use prior biological knowledge about which sites are more likely to be cell-type specific. For instance,

enhancer regions are among the most cell-type-specific regions. Using cell type-specific DNase Hypersensitive Sites information from the NIH Epigenomics Roadmap and ENCODE can in principle also help identify the sites that are more likely to be bonafide cell type-specific DMPs.

2. Ref-free:

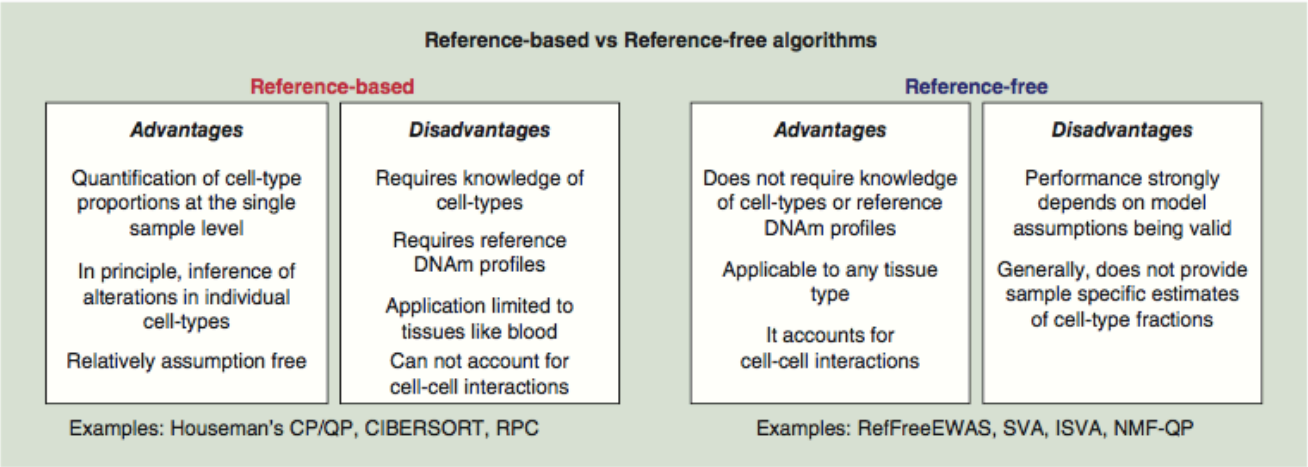

Figure 1. Reference-based versus reference-free inference. Summary of the main advantages and disadvantages of the two major inference-paradigms for adjusting cell-type heterogeneity in EWAS.
DNAm: DNA methylation; EWAS: Epigenome-wide association study.

Cons of ref-based:

1. Even in the case of blood, different cell subsets exit
2. In other complex tissues such as breast or skin, which contain a mixture of widely different cell types, including epithelial, adipose, fibroblasts and immune cell types, these cells are known to interact with each other, which can modify their (expression) profile, so that reference profiles obtained from cell types ex vivo may not be representative of their profiles in vivo. (likely to present in DNAm, too!)

Ref free:

1. EWASher, RefFreeEWAS, ReFACTor
2. SVA, ISVA (confounder can be others instead of cell compostion)
3. RUV
4. NMF (combine NMF with Houseman ref-based, i.e. QP → NMF-QP)

- One key assumption of ReFACTor and EWASher is the top component of variation in data is driven by cell-type composition. (may not be valid!)
- Preferable! RefFreeEWAS, SVA, ISVA use the phenotype of interest from the outset in a supervised framework.
  (In SVA or ISVA, surrogate variables representing potential confounders are identified in the space of residual variation, in other words, in the data obtained after regressing out the effect due to phenotype of interest)
  (likewise, RefFreeEWAS first regresses out the effect of phenotype of interest on the data and then performs SVD on an augmented matrix.

3. Correcting for cell-type heterogeneity: optimal method?

In terms of actual performance, there is currently a lack of studies offering a comprehensive and objective comparison of these algorithms.

- Comparison study 1:
  So far, the most comprehensive analysis has been performed by McGregor, which compared 6 algorithms (Houseman's reference-based CP method, RefFreeEWAS, SVA, ISVA, EWASher and RUV). This study compared performance using extensively simulated data, as well as real EWAS, concluding that no single method performs best across all evaluation scenarios.
  --- SVA is most robust (reason: relatively free of underlying assumptions and also are able to adjust for confounders other than cell-type composition)
  --- but this study didn't consider many real data, only simulated data

- Comparison study 2
  Another study comparing a number of different algorithms (Houseman's reference-based CP, RefFreeEWAS, EWASher and ReFACTor) on simulated as well as real data: Rahmani.
  --- ReFACTor leads to improved power and accuracy
  --- the study is problematic:

- Comparison study 3 – smoking-EWAS
  There are at least 5 to 6 independent EWAS in whole blood, all leading to the same set of smoking-associated DMPs.
  This common set of smoking-associated DMPs can then be used to define a bona fide gold-standard true positive list.

- Aging and Smoking
  It's fairly well accepted that:
  Specific DNAm alterations or those making up Horvath's clock, are not driven by cell type composition.
  Indeed, it would be hard to explain such tissue-independent age-associated DNAm shifts purely in terms of alterations to the proportion of underlying cell types, since these associations have even been observed in highly diverse and purified cell populations, including stem cell populations.

  On the other hand, this may not generalize to other phenotypes.

  For instance, in the context of smoking, a recent study concluded that differential DNAm of a site mapping to GPR15 (a well-known smoking associated EWAS hit) is caused by an expansion of a normal CD3+ T-cell subtype (marked by overexpression of GPR15) in the peripheral blood of smokers.

  However, whether the increased proportion of this T-cell subtype in the blood of smokers is due to an expansion of a normal pre-existing rare T-cell subtype, or whether it is caused by *de novo* DNAm changes to a common T-cell subtype is still unclear, as the study was

underpowered and could not conclusively assess the effect of smoking carcinogens on DNAm patterns.

4. **Blueprint (Resources for blood EWAS)**

   DNA methylomes of all major and minor blood cell subtypes are generated as part of Blueprint.
   Helps generate more complete reference DNAm databases, allowing more refined higher resolution analyses of blood EWAS.

   IHEC samples, WGBS data
   Yet an immediate statistical challenge in translating WGBS methylomes from the International Human Epigenome Consortium (IHEC) and Blueprint into reference DNAm probes that could be used for cell type deconvolution in Illumina 450k/EPIC EWAS is that the coverage of WGBS data is also limited.

   For instance, the overlap between WGBS data for IHEC samples and Illumina 450k probes is typically only around 40% and can range from as low as 15% to as high as 70%.

   *** generating more complete reference DNA methylomes for all major human cell types, as done by IHEC, will also allow direct application of reference-based methods to many other complex tissues like breast, brain, skin or cervix,

   *** optimize reference database: as the number of available reference DNAm profiles grow, there will be a strong need to develop statistical methods which can optimize the construction of reference DNAm databases geared toward analysis of a given tissue type. The collinearity of reference DNAm profiles is detrimental to the quality of inference, and collinearity would increase in line with resolution and number of different cell types present In such reference DNAm databases.

5. **Hierarchical structure inference**

   Given the cell types can be naturally arranged on a developmental tree representing a differentiation hierarchy with human embryonic stem cells at the top and terminally differentiated cells at the bottom, we anticipate that the most powerful cell type deconvolution algorithms may need to incorporate such a hierarchical tree-like structure in the inference.

6. **Single cell**

   Another outstanding question is the extent to which cell-cell interactions can affect and limit the applicability of reference-based methods.

Recent single-cell analyses, performed at the level of gene expression, point toward the importance of cell-cell interactions in modifying specific gene expression programs of tumor, stromal and immune cells.

Upcoming single-cell RNA-seq and DNAm data thus help provide a framework to more objectively compared reference-based with reference-free methods.

Single-cell DNAm analysis also offers a means of improving the resolution of reference-based methods.

7. Across data types

Studies have compared expression (ESTIMATE) and methylation (MethylPurify) based estimates of tumor purity with those obtained using gold-standard SNP/copy-number assays.

** no study yet evaluated the extent to which expression-based and DNAm-based cell type composition estimates agree
**** this is important to assess, since DNAm offers a more stable assay than RNA, yet is not directly functional, rendering a comparison between the two data types highly trivial and interesting.
****** Such comparisons are possible using existing matched DNAm and RNAseq datasets. (TCGA & https://genomebiology.biomedcentral.com/articles/10.1186/s13059-016-1053-6

******** CIBERSORT or CellMix mainly for gene expression, but equally applicable to any other data type, including DNAm.

Integration of multiple data types, while not possible for most EWAS (which by definition only measure DNAm), presents an interesting methodological opportunity to improve the construction of reference DNAm databases, or, as a means of incorporating higher systems-level information as prior biological knowledge, which may be particularly useful for reference-free or semi-reference free approaches. For instance, cell type-specific DMPs which correlate with gene expression *in-cis* may represent a more biologically relevant set of features for constructing reference DNAm databases.

Alternatively, previous studies have shown the value of integrating DNAm data with interaction networks to identify gene modules which mark specific cell types [84]. This may provide another means of adjusting for cell type composition, either in a reference-based or semi-reference free framework

8. Identify which specific cell type drive DMP between phenotypes

In principle, a DMP between phenotypes, which is not driven by the cell type composition effects, could be the result of differential methylation in only one of the underlying cell types, or, at the other extreme, be present in all underlying cell types.

Identifying which specific cell types carry the DNAm alterations is of fundamental importance.

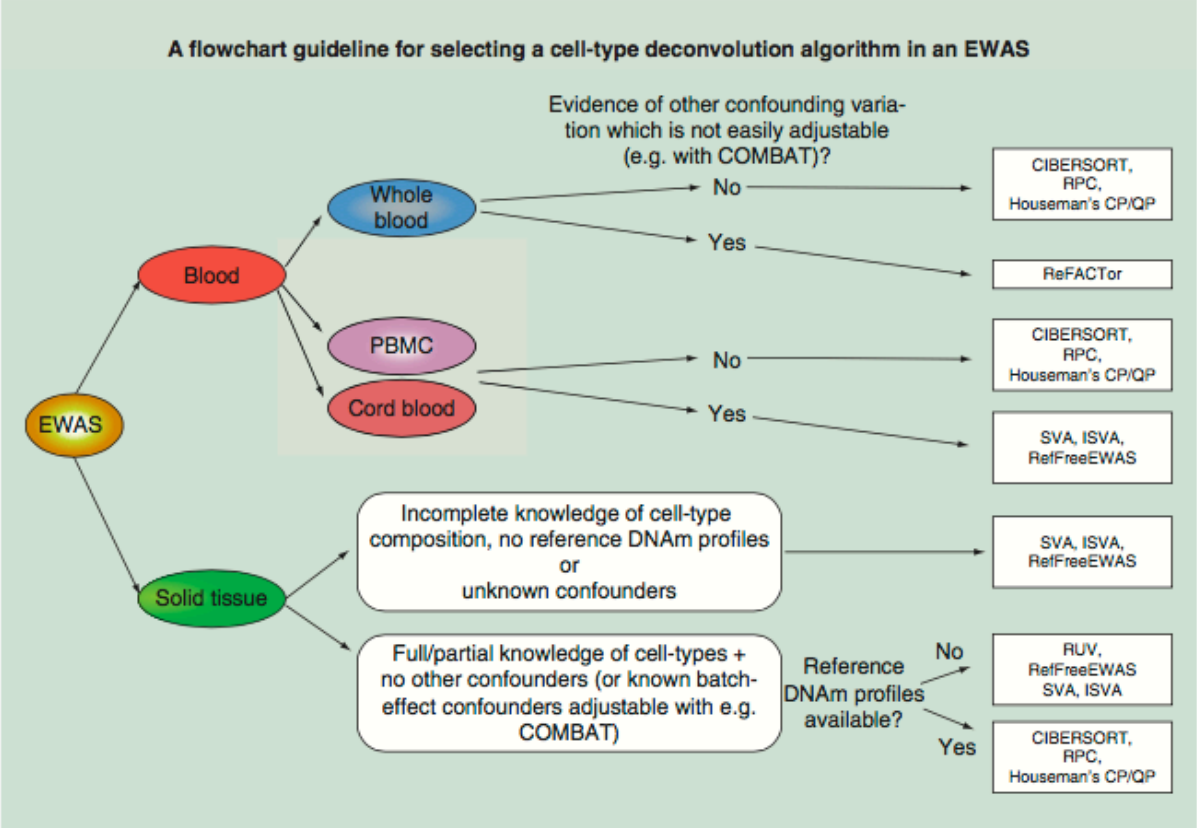

Figure 2. Guideline flowchart for selecting optimal cell-type deconvolution method. Recommended guideline for selecting an optimal algorithm for cell-type deconvolution in an EWAS. In the final column, within each box, recommended algorithms have been ordered in order of preference.
COMBAT: Combatting Batch Effects; CP: Constrained projection; DNAm: DNA methylation; EWAS: Epigenome-wide association studies; ISVA: Independent surrogate variable analysis; QP: Quadratic programming; RPC: Robust partial correlation; RUV: Removing unwanted variation; SVA: Surrogate variable analysis.

Houseman, ref-based:
Houseman, Eugene Andres, William P. Accomando, Devin C. Koestler, Brock C. Christensen, Carmen J. Marsit, Heather H. Nelson, John K. Wiencke, and Karl T. Kelsey. "DNA methylation arrays as surrogate measures of cell mixture distribution." *BMC bioinformatics* 13, no. 1 (2012): 86.

Ref-based, 450k

Jaffe, Andrew E., and Rafael A. Irizarry. "Accounting for cellular heterogeneity is critical in epigenome-wide association studies." *Genome biology* 15, no. 2 (2014): R31

Houseman + NMF (ref-free)

Houseman, E. Andres, Molly L. Kile, David C. Christiani, Tan A. Ince, Karl T. Kelsey, and Carmen J. Marsit. "Reference-free deconvolution of DNA methylation data and mediation by cell composition effects." *BMC bioinformatics* 17, no. 1 (2016): 259.

SVA, ISVA:

Leek, Jeffrey T., and John D. Storey. "Capturing heterogeneity in gene expression studies by surrogate variable analysis." *PLoS genetics* 3, no. 9 (2007): e161.

Teschendorff, Andrew E., Joanna Zhuang, and Martin Widschwendter. "Independent surrogate variable analysis to deconvolve confounding factors in large-scale microarray profiling studies." *Bioinformatics* 27, no. 11 (2011): 1496-1505.

ReFACTor:

Rahmani, Elior, Noah Zaitlen, Yael Baran, Celeste Eng, Donglei Hu, Joshua Galanter, Sam Oh et al. "Sparse PCA corrects for cell type heterogeneity in epigenome-wide association studies." *Nature methods* 13, no. 5 (2016): 443.

ReFreeEWAS:

Houseman, Eugene Andres, John Molitor, and Carmen J. Marsit. "Reference-free cell mixture adjustments in analysis of DNA methylation data." *Bioinformatics* 30, no. 10 (2014): 1431-1439.

Fast-LMM-EASHer (Zou)

Zou, James, Christoph Lippert, David Heckerman, Martin Aryee, and Jennifer Listgarten. "Epigenome-wide association studies without the need for cell-type composition." *Nature methods* 11, no. 3 (2014): 309.

Ref-free, NMF

Moffitt, Richard A., Raoud Marayati, Elizabeth L. Flate, Keith E. Volmar, S. Gabriela Herrera Loeza, Katherine A. Hoadley, Naim U. Rashid et al. "Virtual microdissection identifies distinct tumor-and stroma-specific subtypes of pancreatic ductal adenocarcinoma." *Nature genetics* 47, no. 10 (2015): 1168.

RUV:

Gagnon-Bartsch, Johann A., and Terence P. Speed. "Using control genes to correct for unwanted variation in microarray data." *Biostatistics* 13, no. 3 (2012): 539-552.

Ref-based, blood, brain, breast

Reinius, Lovisa E., Nathalie Acevedo, Maaike Joerink, Göran Pershagen, Sven-Erik Dahlén, Dario Greco, Cilla Söderhäll, Annika Scheynius, and Juha Kere. "Differential DNA methylation in purified human blood cells: implications for cell lineage and studies on disease susceptibility." *PloS one* 7, no. 7 (2012): e41361.

Liu, Yun, Martin J. Aryee, Leonid Padyukov, M. Daniele Fallin, Espen Hesselberg, Arni Runarsson, Lovisa Reinius et al. "Epigenome-wide association data implicate DNA methylation as an intermediary of genetic risk in rheumatoid arthritis." *Nature biotechnology* 31, no. 2 (2013): 142.

Guintivano, Jerry, Martin J. Aryee, and Zachary A. Kaminsky. "A cell epigenotype specific model for the correction of brain cellular heterogeneity bias and its application to age, brain region and major depression." *Epigenetics* 8, no. 3 (2013): 290-302.
Teschendorff, Andrew E., Yang Gao, Allison Jones, Matthias Ruebner, Matthias W. Beckmann, David L. Wachter, Peter A. Fasching, and Martin Widschwendter. "DNA methylation outliers in normal breast tissue identify field defects that are enriched in cancer." *Nature communications* 7 (2016): 10478.

Pan-Cancer

Chakravarthy, Ankur, Andrew Furness, Kroopa Joshi, Ehsan Ghorani, Kirsty Ford, Matthew J. Ward, Emma V. King et al. "Pan-cancer deconvolution of tumour composition using DNA methylation." *Nature communications* 9, no. 1 (2018): 3220.

Cell proportion EWAS & age:

Slieker, Roderick C., Maarten van Iterson, René Luijk, Marian Beekman, Daria V. Zhernakova, Matthijs H. Moed, Hailiang Mei et al. "Age-related accrual of methylomic variability is linked to fundamental ageing mechanisms." *Genome biology* 17, no. 1 (2016): 191.
Yuan, Tian, Yinming Jiao, Simone de Jong, Roel A. Ophoff, Stephan Beck, and Andrew E. Teschendorff. "An integrative multi-scale analysis of the dynamic DNA methylation landscape in aging." *PLoS genetics* 11, no. 2 (2015): e1004996.

Cell proportion EWAS & smoking:

Gao, Xu, Min Jia, Yan Zhang, Lutz Philipp Breitling, and Hermann Brenner. "DNA methylation changes of whole blood cells in response to active smoking exposure in adults: a systematic review of DNA methylation studies." *Clinical epigenetics* 7, no. 1 (2015): 113.
Bauer, Mario, Gunter Linsel, Beate Fink, Kirsten Offenberg, Anne Maria Hahn, Ulrich Sack, Heike Knaack, Markus Eszlinger, and Gunda Herberth. "A varying T cell subtype explains apparent tobacco smoking induced single CpG hypomethylation in whole blood." *Clinical epigenetics* 7, no. 1 (2015): 81.

Comparison:

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